METRIC

FF-B-171/8

30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, FILLING SLOT, ANSI/AFBMA DIMENSION SERIES 03.

The complete requirements, for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

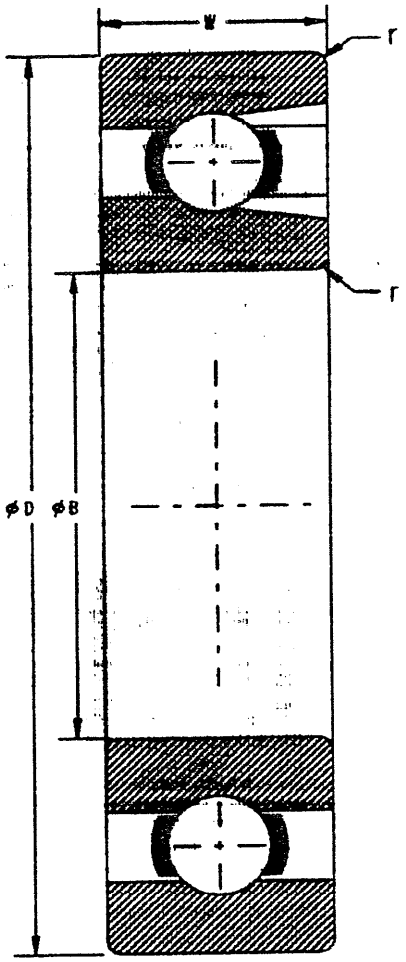


FIGURE 1.
OPEN (NO SHIELDS)
(CODE A)

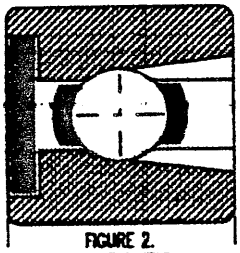


FIGURE 2.
SINGLE SHIELD
OPPOSITE FILLING SLOT SIDE
(NORMAL POSITION)
(CODE B)

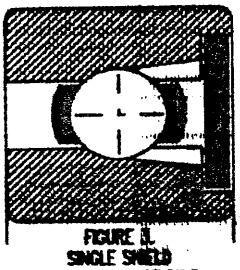


FIGURE 3.
SINGLE SHIELD
ON FILLING SLOT SIDE
(REVERSE POSITION)
(CODE C)

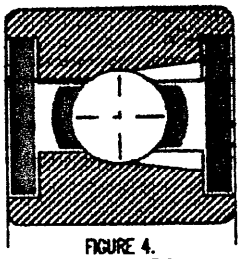


FIGURE 4.
DOUBLE SHIELD
(CODE D)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

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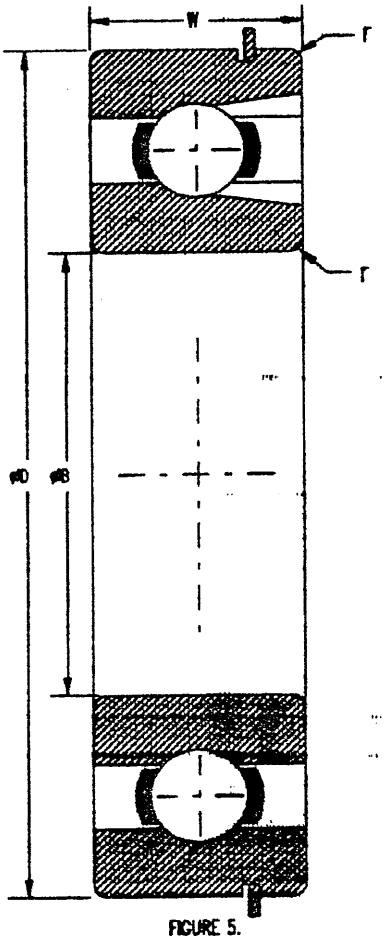


FIGURE 5.
OPEN WITH SNAP RING
ON FILLING SLOT SIDE
(NORMAL POSITION)
(CODE E)

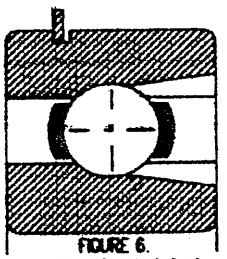


FIGURE 6.
OPEN WITH SNAP RING
OPPOSITE FILLING SLOT SIDE
(REVERSE POSITION)
(CODE F)

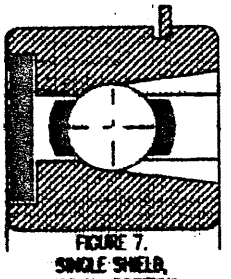


FIGURE 7.
SINGLE SHIELD,
NORMAL POSITION
WITH SNAP RING,
(NORMAL POSITION)
(CODE G)

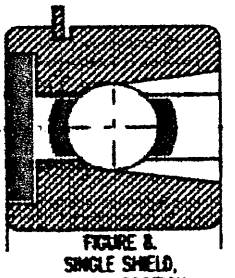


FIGURE 8.
SINGLE SHIELD,
NORMAL POSITION
WITH SNAP RING,
(REVERSE POSITION)
(CODE H)

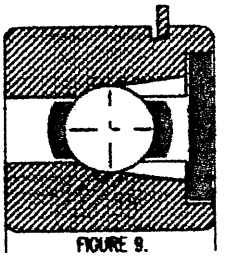


FIGURE 9.
SINGLE SHIELD,
REVERSE POSITION
WITH SNAP RING,
(NORMAL POSITION)
(CODE J)

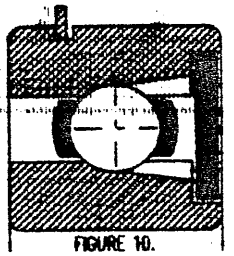


FIGURE 10.
SINGLE SHIELD,
REVERSE POSITION
WITH SNAP RING,
(REVERSE POSITION)
(CODE K)

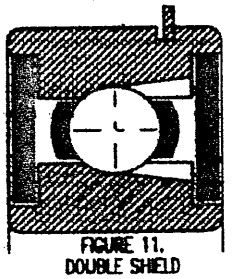


FIGURE 11.
DOUBLE SHIELD
WITH SNAP RING,
(NORMAL POSITION)
(CODE L)

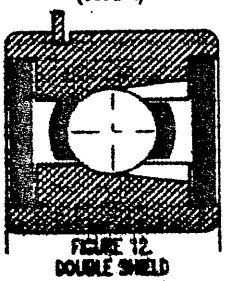


FIGURE 12.
DOUBLE SHIELD
WITH SNAP RING,
(REVERSE POSITION)
(CODE M)

BEARING CONFIGURATIONS - (Continued)

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REQUIREMENTS :

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. **Shields and snap** rings. Shields shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Snap rings shall be fabricated from ferrous material unless otherwise specified arid shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent act. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields and snap rings required shall be specified in the part number by the codes in table III.

4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

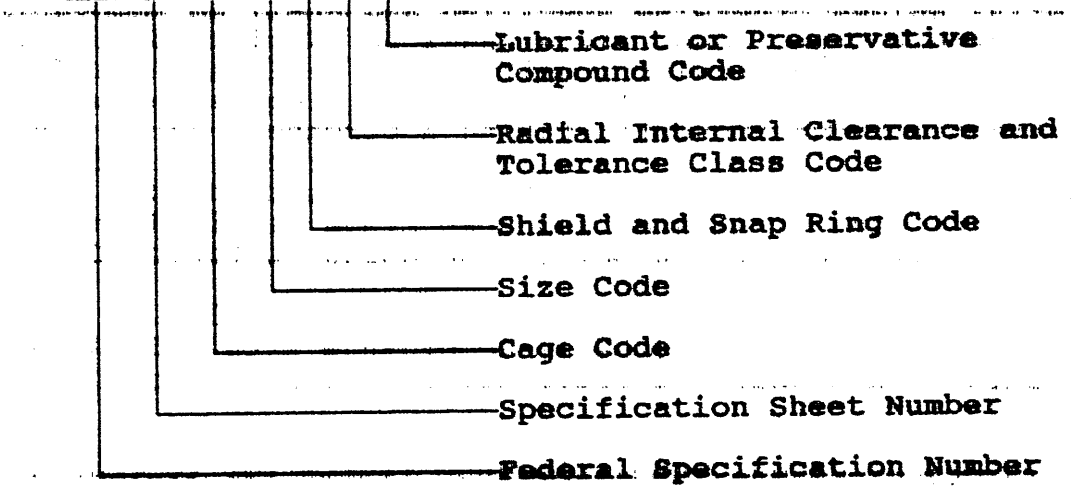
5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. **Lubricant or Preservative Compound.** Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the “C” suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the ‘A” or “B” suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. **Part number.** This part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/08 J 10 K 3 A



FFB171/08J10K3A indicates - Bearing cage-pressed steel or molded plastic; bore 50mm; outside diameter. 110mm width 27mm; single shield, reverse position with snap ring {reverse position); radial internal charge symbol 3 (greater than normal); tolerance class ABEC-1; lubricant in accordance with MIL-G-81322.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
03	17	47	14	1.0	23	3020
04	20	52	15	1.1	27	3500
05	25	62	17	1.1	32	4655
06	30	72	19	1.1	37	6025
07	35	80	21	1.5	43.5	7575
08	40	90	23	1.5	48.5	9270
09	45	100	25	1.5	53.5	11325
10	50	110	27	2.0	60	13270
11	55	120	29	2.0	65	16265
12	60	130	31	2.1	72	18610
13	65	140	33	2.1	77	21090
14	70	150	35	2.1	82	23705
15	75	160	37	2.1	87	25825
16	80	170	39	2.1	92	27990
17	85	180	41	3.0	99	30210
18	90	190	43	3.0	104	32470
19	95	200	45	3.0	109	34250
20	100	215	47	3.0	114	38500
21	105	225	49	3.0	119	41500
22	110	240	50	3.0	124	46500
24	120	260	55	3.0	134	48700

1/ Size codes in Table I are industry bore codes.
2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.
3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

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TABLE III. Codes for Shields and Snap Rings.

Code	Shield and snap ring configuration
A	open (no shields) - fig.1
B	single shield opposite filling slot side, (normal position) - fig. 2
C	single shield on filling slot side, (reverse position) - fig. 3
D	double shield - fig. 4
E	open with snap ring on filling slot side, (normal position) - fig. 5
F	open with snap ring opposite filling slot side, (reverse position) - fig. 6
G	single shield, normal position, with snap ring, (normal position) - fig. 7
H	single shield, normal position, with snap ring, (reverse position) - fig. 8
J	single shield, reverse position, with snap ring, (normal position) - fig. 9
K	single shield, reverse position, with snap ring, (reverse position) - figure 10
L	double shield with snap ring, (normal position)-fig. 11
M	double shield with snap ring, (reverse position)-fig. 12

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

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